Isaac Mubiru

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Kampala, Uganda · EAT (UTC+3), full Berlin-hours overlap · Remote

PROFESSIONAL SUMMARY

Full-stack engineer who has **shipped and maintained production web products across frontend, backend and database** since 2020: **Node.js** on every role, **React** and Tailwind on the client, **TypeScript** APIs on Express. Builds **AI features that have to work in practice**: an OpenAI + Google Vision assistant for prescription OCR over WhatsApp, and an agentic pipeline on Claude Code where a deterministic verification step runs before the model is allowed to judge anything and a human stays on the final submit. **Works with coding agents and reviews their output like a PR.** Led a **zero-downtime migration of 20+ production apps** reporting to the CTO, and turns user needs into revenue features (Express Shipping, +$40 per order). Code: github.com/zac-09

CORE COMPETENCIES

Full-Stack Product Engineering | TypeScript, Node.js & Express | React & Tailwind CSS | AI Features: LLM + Tool Integration | Coding Agents with Output Verification | Data Modelling & Migrations | Background Jobs & Event Pipelines | Ownership from Discussion to Release

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer (US Remote, reports to CTO)

- Led the end-to-end migration of **20+ applications** from Parse/MongoDB to Firebase/GCP with **zero downtime**, reporting directly to the CTO.
- Implemented **Express Shipping**, generating an additional **$40 revenue per order**.
- Built a **3D visualisation feature** for customers using video overlay and ffmpeg, increasing buyer conversion.
- Implemented **real-time two-way sync between MongoDB and Firestore** using Node.js and Google Pub/Sub for message processing.
- Saved the company **$5,000/month** by migrating all services off AWS; migrated file storage from S3 to GCS via a Python script; moved a Webflow site from a Parse backend to Firebase.

Stack: Node.js, JavaScript, React, Firebase/Firestore, MongoDB, GCP Pub/Sub, AWS, Python, ffmpeg, Webflow

MTailor (Contractor)

Mar 2022 – Jul 2022

Backend Migration Contractor (US Remote)

- Built a migration script reading data from MongoDB and writing to Firestore after running complex processing logic.
- Wrote documentation for fellow engineers on the new Firebase SDKs (Node.js and Python) and trained the Ops team on the Firebase dashboard and backup procedures.

Stack: Node.js, JavaScript, MongoDB, Firebase, GCP, AWS

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer (Singapore Remote)

- Extended backend APIs in **TypeScript** using Firebase Cloud Functions and Express.js.
- Added responsive UI pages to a real-time stock-price PWA using **React, Tailwind** and HTML.
- Built ad-hoc jobs querying over **2 million Firestore records** to keep customer prices current from Morningstar APIs.

Stack: TypeScript, Node.js, Express.js, React, Tailwind, Firebase

Mind2matter

Jan 2022 – Jul 2022

Software Engineer (US Remote, reports to CTO)

- Delivered React UIs from Figma designs under tight timelines for a US agency with demanding clients.
- Built backends for DeFi applications using Web3 and Node.js.

Stack: Node.js, JavaScript, React, Tailwind, Firebase, Web3

CodeBits

Jan 2020 – Jul 2021

Software Engineer / Team Lead (Uganda Remote)

- Led a team of 4 developers building legal-tech systems for NGOs: a case management system for FIDA Uganda and a paralegal database for LASPNET Uganda.

- Developed a cross-platform mobile app with **React Native and Expo** (iOS & Android) with Firebase push notifications.
- Built a USSD service for legal aid providers talking to a Node.js backend over gRPC; architected a microservices backend with Apache Kafka, Docker and Kubernetes.

Stack: Node.js, JavaScript, React Native, Expo, Apache Kafka, Docker, Kubernetes, gRPC, Firebase

PROJECTS

Healthcare WhatsApp Assistant with OCR

LLM + vision tools in production

Conversational assistant that reads prescription photos and answers through the WhatsApp Business API. OpenAI for the dialogue layer, Google Vision for OCR, with dialogue state, media ingestion and platform limits handled in the Node.js backend.

Node.js, OpenAI API, Google Vision, WhatsApp Business API

Career-ops — Agentic Pipeline on Claude Code

Coding agents, verified output

Node.js automation where agents scan, fetch and score job postings. A headless-browser check confirms each source is live before any model scores it; scores must cite the source's own wording; dedup history blocks re-grading; agent-written heuristics are reviewed and tightened like any PR; a human makes the final submit.

Node.js, Claude Code, Playwright, Puppeteer, YAML

Parse → Firebase Platform Migration (MTailor)

20+ apps, zero downtime

Owned from plan through cutover: migration tooling, live two-way sync on Pub/Sub, per-app cutover, SDK docs and Ops training so the team could run the new platform. $5,000/month saved on completion.

Node.js, Python, MongoDB, Firestore, GCP Pub/Sub

EDUCATION

B.Sc. Computer Engineering — Makerere University — Aug 2018 – Sep 2022

Kampala, Uganda

SKILLS

Proficient: Node.js, TypeScript, JavaScript, React, Firebase, GCP, Python, MongoDB, SQL, Git, Linux, HTML, CSS
Frontend: React, Tailwind CSS, React Native, Expo, PWA
AI: OpenAI API, Google Vision, Claude Code, agentic pipelines with browser verification
Backend & infra: Express.js, Cloud Functions, Google Pub/Sub, Apache Kafka, gRPC, Docker, Kubernetes, AWS
Intermediate: Go, Rust
Languages: English (fluent; 5+ years US/Singapore remote), Kiswahili